

Data co supply chain project dataset report

Demand Forecasting & S&OP Analytics Report

DataCo Smart Supply Chain — Power BI Dashboard Project

Prepared by: Abishek Kolanjiappan OSCM & Marketing, University of Toledo

Dataset: DataCo Smart Supply Chain (Kaggle), 180,519 orders, January 2015 – September 2017

Overview

This powerbi project has built by analysing 180,000 (rows) orders from a global retailer to answer three main question .How accurate is demand forecast ,where are we losing money on delivery and what is actually driving the sales trends

I have built proper star schema from scratch and wrote about 20 DAX measures covering from basic to KPI's rolling 3month average forecast model and headline finding was that our late delivery problem is not due to regional logistics issue -it was a systemic scheduling problem - It was systemic scheduling problem-it showed up almost similar identically across 23 locations

1.Methodology

Data preparation

The raw data contains about (180,519) rows was cleaned by removing some unwanted data like (customer email customer ,product description etc)

Data Modelling

So rather than relying on one data set I created a dimension table based on date with star schema . A custom dimension table was created spanning a year from January 1st 2015 though approximately September 2017 which helps in time intelligence calculation. While doing this I noticed the mismatched data value problem so I used **DATEVALUE**. Ensuring model function correctly

Measures development

20+ measure have been created and then it was used for calculating Core KPI, time intelligence,forecast

Forecast Approach

A 3 month average was used for forecasting average on baseline forecasting model. A ideal choice to establish defensive benchmark before considering more complex it was measured by MAPE (mean absolute percentage error)

2.Key Findings

2.1 Forecast Accuracy

The 3-month moving average model produced an overall MAPE of 5.53%. Performance was consistent across most of the analyzed period, with one notable exception: September 2017 recorded a 10.35% forecast miss, roughly double the average error

2.2 Delivery Performance

METRICS	VALUE
Late Delivery Rate	54.41%
Avg. Actual Shipping Days	3.50 days
Avg. Scheduled Shipping Days	2.94 days
Shipping Delay (Actual – Scheduled)	0.56 days

At first glance 54% , Late rate suggest significant failure in operational failure. However regional level analysis shows about late rate fall with a narrow of 48-56% across all 23 region. Combined with the modest 0.56-day average delay, this points to a structural issue with how delivery windows are promised, rather than inconsistent fulfillment execution

2.3 Sales Trend & Growth

Month over month and year over year growth measures were built to track momentum beyond raw totals. These metrics were used to analyse whether the growth is picking up growing up/down or stable. Supporting the supply chain operations

3. Recommendations

To fix the delivery delivery rate across all region the 1st step is review of how scheduled shipping time are set rather than involving in logistic operations.I would suggest if there is any seasonal patterns are confirmed a method such as exponential smoothing or ARIMA model should be tested against 3 month average

4. Dashboard overview

The analysis is delivered through a 4-page interactive Power BI dashboard:

Executive Summary — top-line KPIs, sales trend, category and regional breakdowns

Forecast vs. Actual — forecast accuracy tracking and monthly detail

Sales Trend & Growth — month-over-month and year-over-year growth analysis

Delivery & Fulfillment — shipping performance and regional delivery comparison